

Business Problem Analysis

TeleConnect Analytics Hub

1. Company Background

The telecommunications industry is one of the most competitive service sectors, where customer satisfaction, service quality, and customer retention directly influence long-term business growth and profitability. Telecom providers manage millions of customer interactions every day through subscription plans, billing systems, customer support channels, and network services. These interactions generate vast amounts of operational data that contain valuable insights into customer behavior, revenue performance, service quality, and business operations.

Despite the availability of large datasets, many organizations struggle to convert raw operational data into actionable business intelligence. Business teams often rely on manually prepared reports generated from multiple systems, making it difficult to gain a unified view of organizational performance. This limits the organization's ability to identify business risks, monitor key performance indicators, and make timely strategic decisions.

To remain competitive, telecom organizations require centralized analytical solutions that provide accurate, real-time insights into customer trends, revenue performance, operational efficiency, and service quality.

2. Business Challenges

The organization faced several operational and analytical challenges that reduced the effectiveness of business decision-making.

Limited Visibility into Customer Churn

Management lacked a centralized mechanism to monitor customer churn across different customer segments, subscription plans, and geographic regions. As a result, identifying high-risk customers required significant manual effort, delaying retention initiatives.

Fragmented Business Reporting

Business reports were generated from multiple data sources and departments without a unified reporting framework. This resulted in inconsistent reporting, duplicated analysis, and increased effort for management teams.

Revenue Leakage

The organization could measure overall revenue but had limited visibility into revenue lost due to customer churn. Without understanding the financial impact of churn, prioritizing customer retention investments became challenging.

Inefficient Complaint Monitoring

Customer complaints were stored operationally but were not analyzed strategically. Management could not easily determine complaint trends, identify recurring service issues, or evaluate how complaints influenced customer satisfaction and churn.

Limited Customer Segmentation

The business lacked detailed analysis of customer demographics, customer types, subscription plans, tenure, and regional distribution. This restricted the organization's ability to develop targeted marketing campaigns and personalized retention strategies.

Slow Decision-Making

Preparing business reports required manual consolidation of multiple datasets. Decision-makers often worked with outdated information, reducing the speed and effectiveness of operational and strategic planning.

3. Existing Pain Points

The absence of a centralized Business Intelligence solution created several operational limitations throughout the organization.

- Business managers were unable to monitor critical KPIs from a single platform.
- Revenue teams lacked detailed insights into subscription plan performance and customer revenue contribution.
- Customer service managers could not easily identify complaint patterns or evaluate their impact on customer satisfaction.
- Executive leadership had limited visibility into churn trends, revenue loss, and customer retention performance.
- Operational teams spent considerable time preparing reports manually instead of focusing on strategic analysis.

These limitations reduced operational efficiency and prevented stakeholders from making timely, evidence-based decisions.

4. Stakeholder Requirements

Executive Management

- Monitor overall business performance using executive KPIs.
- Track customer growth, churn, retention, and revenue trends.
- Evaluate organizational performance through interactive dashboards.
- Support strategic planning with data-driven insights.

Sales and Marketing Team

- Identify profitable customer segments.
- Analyze subscription plan performance.
- Understand customer demographics and regional distribution.

- Support customer acquisition and retention strategies.

Customer Service Department

- Monitor complaint volume and complaint categories.
- Evaluate customer satisfaction.
- Identify service quality issues affecting customer experience.
- Support service improvement initiatives.

Finance Team

- Measure total revenue and revenue loss.
- Analyze financial impact of customer churn.
- Evaluate revenue contribution by customer type and subscription plan.
- Monitor business profitability using financial KPIs.

Business Analysts

- Access centralized, reliable business data.
- Perform interactive analysis using filters and drill-down capabilities.
- Generate insights without manual report preparation.
- Support business recommendations through data visualization.

5. Business Questions

The project was designed to answer several critical business questions that directly support organizational decision-making.

- What is the current customer churn rate?
- Which customer segments have the highest churn risk?
- Which subscription plans generate the highest revenue?
- How much revenue is lost due to customer churn?
- Which customer groups contribute the highest business value?
- Which geographical regions require customer retention initiatives?
- What are the most common customer complaints?
- How does customer satisfaction influence retention?
- Which customer segments should be prioritized for marketing campaigns?
- Which operational areas require immediate management attention?

These questions guided the development of the dashboards, KPIs, SQL queries, and analytical reports throughout the project.

6. Proposed Business Solution

To address the identified business challenges, TeleConnect Analytics Hub was developed as a centralized Business Intelligence platform that integrates customer, revenue, complaint, and retention analytics into a single reporting solution.

The solution consolidates data from operational sources, performs data transformation and KPI calculations, and presents business insights through interactive Power BI dashboards supported by SQL and DAX.

By combining descriptive analytics with business-focused visualizations, the solution reduces manual reporting effort while improving the quality and speed of decision-making.

7. Project Goals

- Develop a centralized Business Intelligence reporting solution.
- Improve visibility into customer behavior and churn.
- Enable continuous monitoring of business KPIs.
- Analyze customer demographics and subscription performance.
- Measure revenue performance and revenue loss.
- Evaluate customer complaints and service quality.
- Support customer retention initiatives through data analysis.
- Improve reporting efficiency through interactive dashboards.
- Enable faster, evidence-based business decision-making.

8. Expected Business Outcomes

The implementation of TeleConnect Analytics Hub is expected to provide measurable business benefits across multiple departments.

The organization will gain a centralized reporting platform that improves visibility into customer performance, operational efficiency, and financial performance.

Business users will spend significantly less time preparing reports manually, allowing greater focus on strategic analysis and business improvement initiatives.

Management will be able to identify customer churn trends earlier, monitor revenue performance more effectively, evaluate service quality continuously, and prioritize initiatives that improve customer satisfaction and retention.

Overall, the project enables a more data-driven decision-making culture by providing accurate, interactive, and business-focused insights that support sustainable organizational growth.